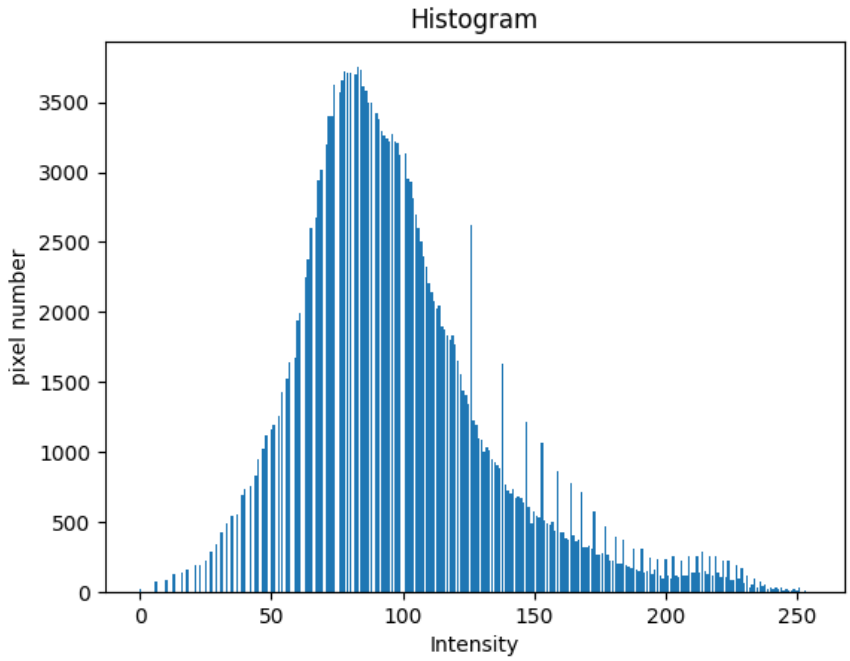
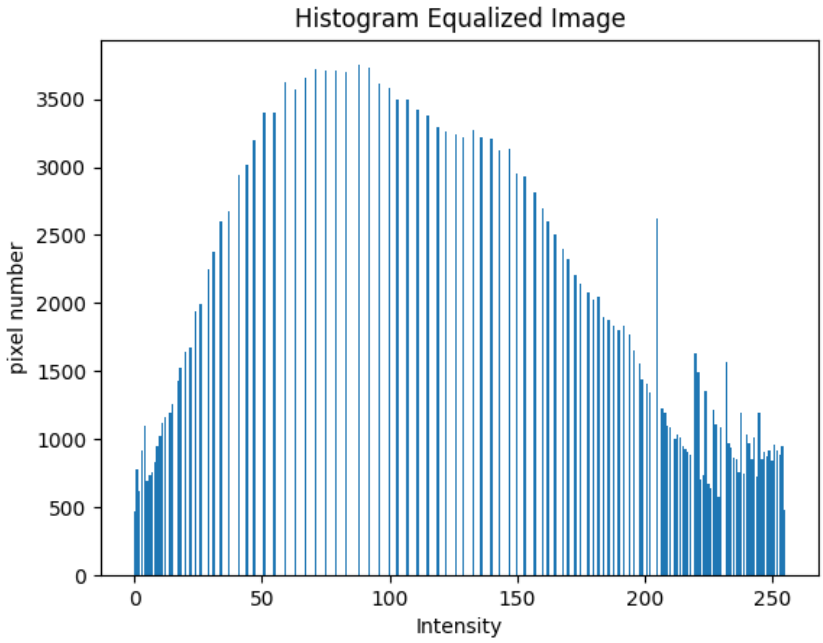
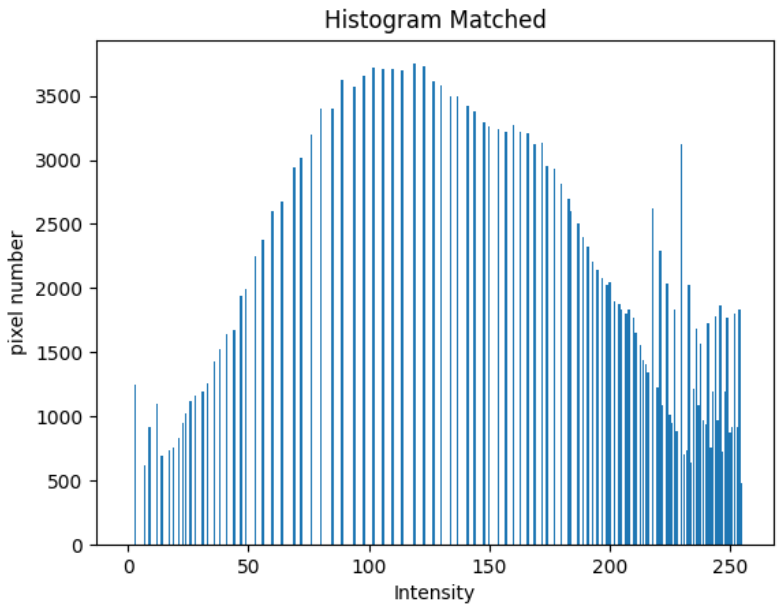
1. Original Image



2. Histogram Equalization



3. Histogram Matching



4.

1) We can see the original image is relatively dark, as its histogram is skewed toward the left, which is the lower intensity region. Also, areas of the image corresponding to the same building usually have similar tones, resulting in segments of the histogram with more pronounced pixel intensity distributions.

2) The histogram becomes more uniform, and the overall brightness of the image appears significantly increased to the naked eye. However, the resolution seems to decrease somewhat, causing the histogram to have even more pronounced segmentation.

3) The image's histogram is adjusted to fit a specified target distribution function:

$$p_z(z_q) = c * {z_q}^{0.4}$$

Using CDF, we can calculate c:

$$\sum_{q=0}^{255} c * {z_q}^{0.4}\, \mathrm{d}z_q = 1$$

$$\mathrm{c} = \frac{1}{1675.54} = 0.0006$$

After matching, the histogram is skewed more toward higher pixel values compared to equalization, producing an effect that looks somewhat overexposed.